BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Defensive Design, Validation and Testing

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

BLACK PEN: Answer all four before turning to the answers.

1. What does assignment do to a variable?

2. What does the MDR hold?

A) Data moving to/from memory
B) The next instruction's address
C) Only ALU calculation results
D) Only user account names

3. What does the Program Counter hold?

4. Similar CPUs have different clock speeds and cores. Some tasks can be shared across cores. Correct both claims. Explain why. (a) Double clock speed always halves run time. (b) Each extra core doubles every program's speed.

Today's objectives

Explain defensive design and input validation. Distinguish validation from authentication.

Key vocabulary - Read, cover, recall

defensive design
Designing programs to handle invalid input and misuse.

validation
Checking entered data against defined rules.

authentication
Checking a user's identity.

boundary data
Valid data at the extreme limits of an allowed range.

Write one definition in your own words:

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Learn and practise

New learning - Four key ideas

DEFENSIVE DESIGN	Anticipate unsuitable input; handle it safely.
VALIDATION	Check data against a type, range or length rule.
AUTHENTICATION	Check an identity. This is not validation.
TESTING	Try normal, boundary, invalid and erroneous data.

MARK VALIDATION - ACCEPT 0 TO 100 INCLUSIVE

-1	0	56	100	101
Reject	Accept	Accept	Accept	Reject

'hello' is erroneous (wrong type); it must be handled safely.

I Do - Worked example

A program accepts a mark from 0 to 100 inclusive. Choose test data and explain the expected results.

1. The valid mark range is 0 to 100 inclusive.
2. Normal: 56 -> accept. Boundaries: 0 and 100 -> accept.
3. Invalid numeric values: -1 and 101 -> reject.
4. Erroneous type: 'hello' -> reject or handle safely.

We Do - Guided practice

Classify 100 and 101 for a 0-100 mark rule.

BLACK PEN - complete your explanation:

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Predict the number printed when one is added.

```
value = 255
print(value + 1)
```

BLACK PEN: complete the ONE short task on paper first.

TEST by tracing. IMPROVE by checking the full model on Page 4.

You Do - Independent exam questions

Q2. Define input validation. [2 marks]

Q3. Give two boundary values and one invalid value for marks from 0 to 100 inclusive. [3 marks]

Q4. Distinguish authentication and validation; explain why iterative testing is useful. [4 marks]

Exit Ticket - True or False?

Answer before checking Page 4.

1. Validation and authentication mean exactly the same thing. ☐ T ☐ F
2. Validation checks data against a rule. ☐ T ☐ F
3. An invalid mark of 101 must be accepted in the range 0-100. ☐ T ☐ F

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: Mark fairly. Tick correct points and fix any mistakes.

Use the mark allocations; then rewrite one answer in full.

Do Now - Answers

Q1 It stores a value in the named variable, replacing its previous value if one existed. ☐

Q2 A - Data moving to/from memory ☐

Q3 The address of the next instruction to be fetched. ☐

Q4 (a) Higher clock speed means more cycles per second, which may process more instructions. (b) More cores help when work can be shared between them. Speed also depends on the task and other CPU features. ☐

Code First - Model

```
value = 255
print(value + 1)
```

EXPECTED: 256 ☐

We Do - Check the model

100 is valid upper boundary; 101 is invalid.

Exit Ticket - Check

1. FALSE 2. TRUE 3. FALSE

Explain one false statement using the key ideas.

You Do - Self-mark

Q2 Check input against rules [1]; reject unsuitable data [1]. ☐

Q3 0 and 100: valid boundaries [2]; 101 or -1: invalid [1]. ☐

Q4 Authentication checks identity [1]; validation checks data [1]; iterative tests find faults early [2]. ☐

Try the question again

1. Copy the FULL question you got wrong (or found hardest):

2. In GREEN PEN, write a NEW answer. Include working if needed: